

1) Image Formation

There are 3 types of images:

- **Grayscale:** $f(x, y) \in [0, 1]$ continuous function that represents the intensity of the gray level
- **Color Image:** $f_c(x, y, \lambda)$ where λ is the value for each wavelength. A good approximation is to use 3 continuous functions for each pixel color: $f_{R/G/B}(x, y)$ (details later)
- **Digital Images:** as before but $x, y \in \mathbb{N}$. So the values of f are discrete and finite

This course deals with digital images that are stored as 2D matrices with 8-bit values per pixel $[0, 255]$.

How are images formed?

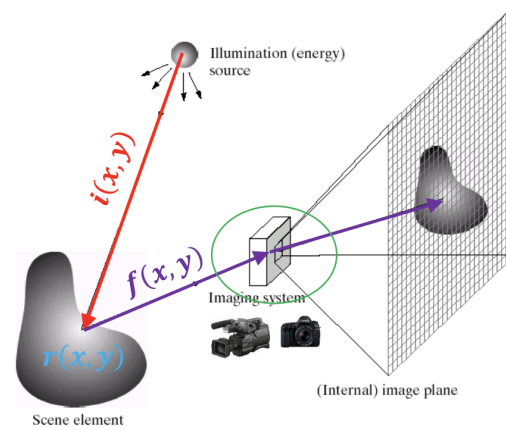
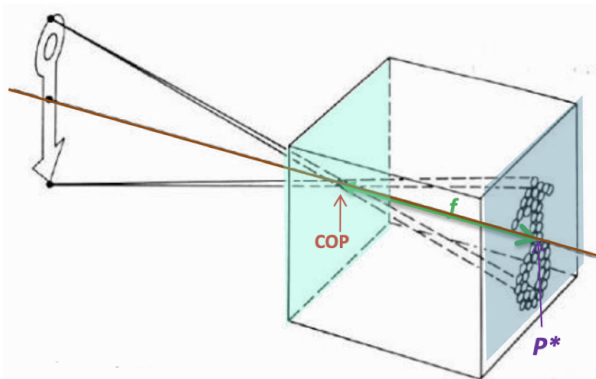


Image Formation

$f(x, y) = i(x, y)r(x, y)$ this shows that the final image will be the light source (i) multiplied by the reflectance (r) of the objects. The reflected light is the one reaching the sensor.

But the light gets scattered everywhere and therefore we need a barrier (pinhole) to block most duplicate rays. This produces an inversion in the result.



Pinhole Camera Model

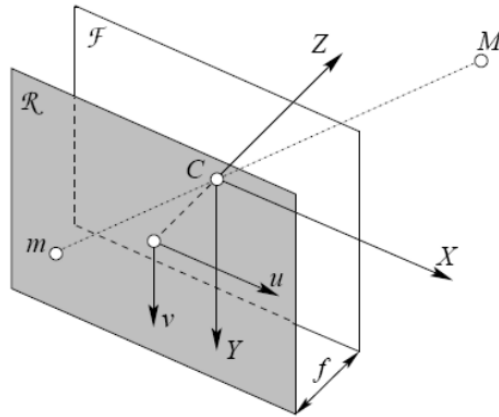
The pinhole is called **Center of Projection** (COP, C) and the perpendicular line passing through it is the **optical axis** this plane is called focal plane (F). The image is built on the **imaging plane** (R) and the **principle point** (P*) is the intersection between R and optical axis. The **focal distance** (f) is the distance between C and R.

This creates a projection from 3D space $\tilde{M} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ to a 2D space $\tilde{m} = \begin{bmatrix} x \\ y \end{bmatrix}$. In this model we have that $u = f \frac{x}{z}, v = f \frac{y}{z}$, which is clearly non linear due to the division by z which is the perspective shift.

By adding an extra dimension $M = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$, $m = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$ and by normalizing by the last coordinate ($\tilde{m} = \alpha m$) we have

homogeneous coordinates, this allows for a linear mapping, that is

$$\lambda m = PM$$



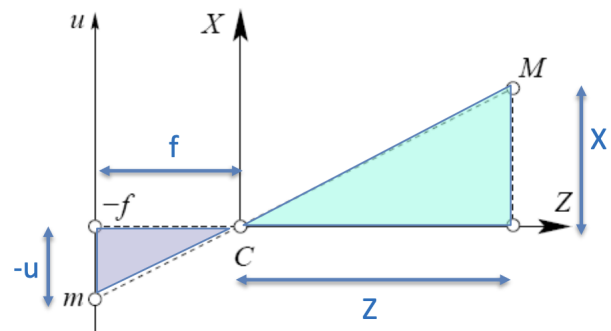
Homogeneous Coordinates

1.1) Mathematical Model for 3D to 2D Projection

The mathematical model must map the **world reference coordinates** ($\{x, y, z, (1)\}$) to the **camera reference system** ($\{u, v, (1)\}$).

Simplified Model

- Notice that optical axis is the z axis
- Set $C = (0, 0, 0)$ and $P^* = (0, 0)$
- (u, v) aligned with (X, Y)



Graphical View

from here:

$$\begin{cases} u = \frac{-f}{Z}X \\ v = \frac{-f}{Z}Y \end{cases} \xrightarrow{\text{hom. coord}} Z \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = Z \begin{bmatrix} \frac{-fX}{Z} \\ \frac{-fY}{Z} \\ 1 \end{bmatrix} = \begin{bmatrix} -fX \\ -fY \\ Z \end{bmatrix} = \begin{bmatrix} -f & 0 & 0 & 0 \\ 0 & -f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

and we can see that

$$m \approx PM$$

with P the 3x4 matrix. Note how the image is reversed ($-f$).

A more general model:

Forst we must **drop some assumptions**:

- Arbitrary camera position and orientation ($\theta \approx 90^\circ$ ignored for this course)
- (optional) real lenses have distortion
Define the **intrinsic parameters**
- Focal Length (f)
- Pixel Size in u, v directions (p_u, p_v) and number of pixels ($k_u = 1/p_u, k_v = 1/p_v$)
- Position of principal point (u_0, v_0)
And the **extrinsic parameters**
- Camera position
- Viewing Direction

First define the 3x3 matrix V :

$$V = \begin{bmatrix} -k_u & 0 & u_0 \\ 0 & -k_v & v_0 \\ 0 & 0 & 1 \end{bmatrix}$$

And P will become the matrix multiplication between V and the P of the simplified model:

$$P = \begin{bmatrix} -k_u & 0 & u_0 \\ 0 & -k_v & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -f & 0 & 0 & 0 \\ 0 & -f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} -fk_u & 0 & u_0 & 0 \\ 0 & -fk_v & v_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

This matrix contains the **K matrix, the intrinsic parameter matrix** that is the P without the last 0 column and contains 4 (+1 skew) **intrinsic parameters**

- $\alpha_u = fk_u = f/p_u, \alpha_v = fk_v = f/p_v$ (focal length of pixels)
- u_0, v_0
- (θ)

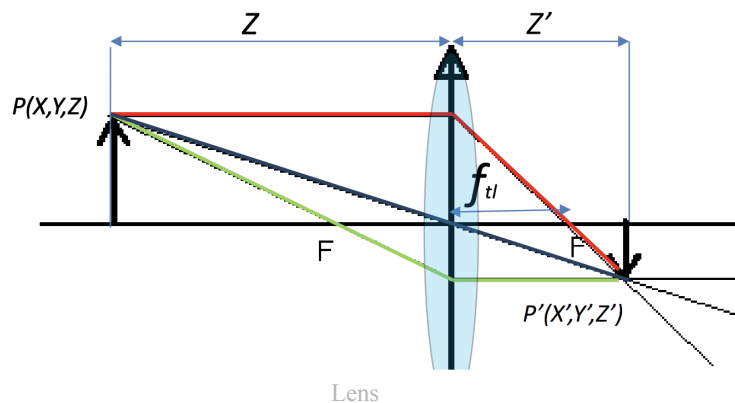
✓ Todo

Extrinsic Parameters → Full model

Finally the model has 11 degrees of freedom (5 intrinsic + 3 rotation + 3 translation).

The final image will lose some information like the correct perspective, angles, dimension and distance. However straight lines will remain straight. Notice that due to the angle loss parallel lines are not parallel anymore.

1.2) Real Cameras



The lens corrects the path of the rays.

- Any ray parallel to the axis proceeds towards F

- Any ray passing through the center is unchanged
- Any ray passing through F before the lens becomes parallel

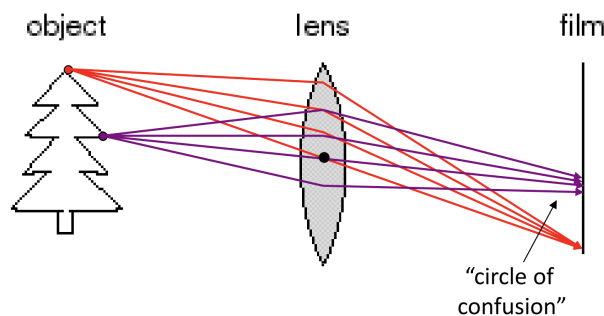
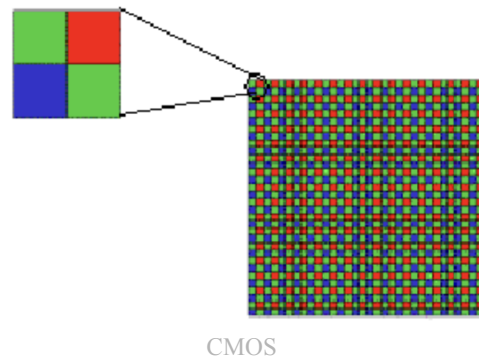
If an object at distance Z is at focus ad distance Z' then the following relation holds:

$$\frac{1}{f_{tl}} = \frac{1}{Z} + \frac{1}{Z'}$$

If the distance is different from Z' we enter the **circle of confusion** where the rays don't focus on the plane. If this circle is bigger than the pixel (cmos sensor) then we get blurring.

A lens allow to get more light coming in.

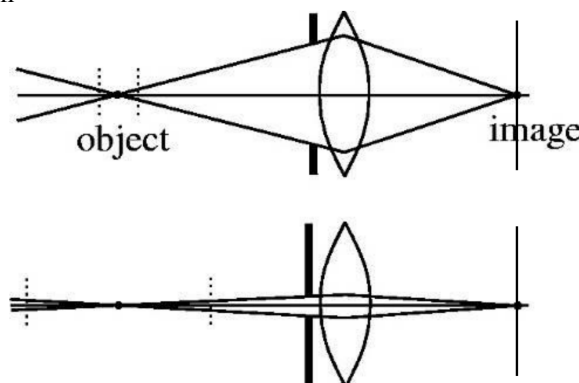
A digital camera is made of a grid of CMOS sensors. Each sensor reacts to each photon that arrived. They **do not sense wavelength**, to get colour a filter is used. E single pixel is made of 1 red 2 green and 1 blue filter on each diode (**Bayer pattern**). By interpolating the values the RGB is found



Example of Out Of Focus

Aperture and shutter speed regulate exposure, that is, how much light enters the sensor. Higher shutter speed allows for more light and therefore more motion blur.

- Shutter speed regulates the time the sensor is open in fractions of a second: $1/x$
- Aperture regulates the diameter of the pinhole expressed by focal length divided by number: $a = f/N$ it increases with $N = 2$ as a sqrt of 2: $2 \rightarrow 2\sqrt{2} = 2.8 \rightarrow 2.8\sqrt{2} = 4 \rightarrow \dots$ This is proportional to the area and $\sqrt{2}$ doubles the light coming in



Reciprocity: The same exposure is obtained with and exposure twice as long and an aperture area half as big

$$a_1 f_1 = a_2 f_2 \rightarrow \begin{cases} a_2 = a_1/2 \\ f_2 = 2f_1 \end{cases}$$

1.3) Colorimetry

2) Intro to Image Processing

Recall the difference between an analog and a digital image. How do we discretize the values? By sampling and quantizing!

- Sampling:

$$f(m, n) \stackrel{\Delta}{=} f(m\Delta x, n\Delta y)$$

where $\Delta x, \Delta y$ correspond to the pixel size and are the sampling period.

- Quantization:

$$f_d(m, n) = Q[f(m, n)]$$

A quantizer has $L = 2^k$ levels. Each $M \times N$ image requires $b = MNk$ bits to store the data.

Remark.

This is greyscale, for the colors you do this three times with each rgb value. An rgb image requires thrice the space $b_{rgb} = 3b = 3MNk$

Images are 2D signals and therefore the frequency analysis has to be performed also in 2 dimensions.

$$F(u, v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) e^{-j2\pi(ux+vy)} dx dy$$
$$f(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(u, v) e^{j2\pi(ux+vy)} du dv$$

But most importantly the discrete transforms:

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$
$$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

However in the DFT **the transforms are discrete and periodic**

Theorem 1 (Fourier Transform Separability).

A 2D DFT can be computed by M 1D-DFT (one) for each column and N 1D DFT (one) for each row.

$$F(u, v) = \sum_{x=0}^{M-1} e^{-j2\pi\frac{ux}{M}} \sum_{y=0}^{N-1} F_{1D \text{ on } y \text{ axis}}(u, v) = \sum_{x=0}^{M-1} F'(x, v) e^{-j2\pi\frac{ux}{M}}$$

Where $F'(x, v)$ is the sum of all 1D DFT done on the Y axis.

Theorem 2 (Separability Theorem).

If the system allows the decomposition $h(x, y) = f(x)g(y)$ then the 2D DFT will be $H(u, v) = F(u)G(v)$

Remark.

Do not confuse the separability thm with product $\leftrightarrow$ convolution property! In fact, in general if

$$h(x, y) = f(x, y)g(x, y) \xrightarrow{\mathcal{F}} H(u, v) = F(u, v) * G(u, v)$$

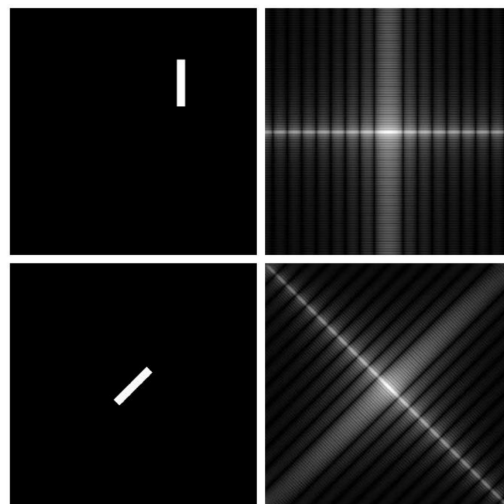
Quick Recap on DFT Properties**Polar Representation:**

$$F(u, v) = |F(u, v)|e^{j\phi(u, v)}$$

Spectrum (Magnitude):

$$|F(u, v)| = \sqrt{\Re[F(u, v)] + \Im[F(u, v)]}$$

Notice how a translation in the image doesn't change the spectrum, however a rotation changes the frequency (it rotates the spectrum)

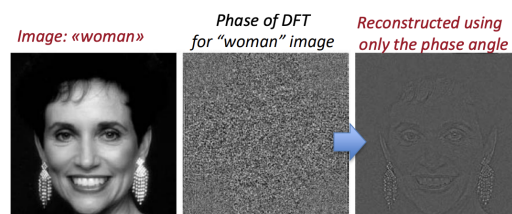


Translation and Rotation

Phase:

$$\phi(u, v) = \arctan \left(\frac{\Re[F(u, v)]}{\Im[F(u, v)]} \right)$$

The phase contains the most info on the image



Example

Convolution/Product:

$$f(x, y) * h(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n) h(x - m, y - n) \xrightarrow{\mathcal{F}} F(u, v) H(u, v)$$

See also this [Remark](#).

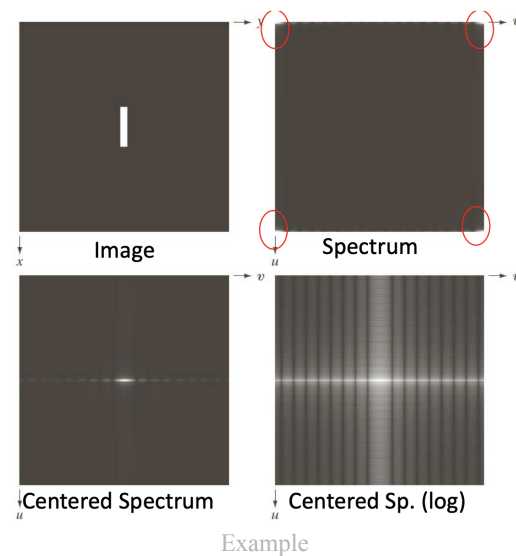
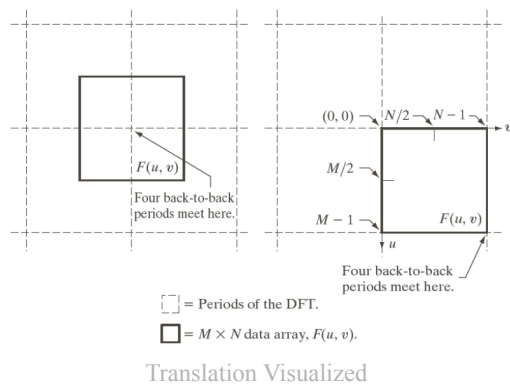
Separability:

See [Theorem 2 \(Separability Theorem\)](#).

Translation To Center of The Frequency Rectangle:

$$f(x - M/2, y - N/2) \xrightarrow{\mathcal{F}} F(u, v)(-1)^{u+v}$$

In particular the power is concentrated in the lower frequencies (start and end of period), the shift allows to have the energy in the middle region.



Differentiation:

Gaussian: (dft of gaussian is gaussian)

$$A 2\pi\sigma^2 e^{-2\pi^2\sigma^2(t^2+z^2)} \xrightarrow{\mathcal{F}} A e^{-(\mu^2+v^2)/2\sigma^2}$$

Band-limited Images:

Theorem 3 (Band-Limited).

A 2D signal is said to be strictly band limited if its fourier transform is such that

$$F(u, v) = 0 \text{ for } |u| > u_{\max}$$

and

$$F(u, v) = 0 \text{ for } |v| > v_{\max}$$

2D Sampling Theorem

Theorem 4 (2D Sampling Theorem).

Let $f(x, y)$ be a band limited image uniformly sampled over an orthogonal lattice with spacing $\Delta X, \Delta Y$. Then, if the sampling rate is greater than the nyquist rate, $f(x, y)$ can be reconstructed from the sampled values without any loss of information, that is

$$F_x = \frac{1}{\Delta X} > 2u_{\max} \text{ and } F_y = \frac{1}{\Delta Y} > 2v_{\max} \implies F^{-1}(u, v) = f(x, y)$$

2.2) Geometric (Affine) Transforms**2.3) Single-Pixel Operations**

It is possible to modify each pixel on a one by one basis to change the output image. For example an color inversion is obtained by mapping

$$s = T(r) = (L - 1) - r$$

Or the gamma transformation:

$$s = cr^\gamma$$

where $\gamma > 1$ images get darker and $\gamma < 1$ images get brighter.

The dynamic range can be adjusted via a **histogram tuning** approach:

- First compute the Probability Density Function (PDF) as

$$p_r(r_k) = \frac{n_k}{MN}$$

where n_k is the number of pixels with intensity r_k

- Map the values to the CDF

$$s_k = T(r_k) = (L - 1) \sum_{i=0}^k p_r(r_i) = \frac{L - 1}{MN} \sum_{i=0}^k n_i$$

- Compute the PDF and CDF of the target function so that

$$G(z_q) = s_k \rightarrow z_q = G^{-1}(s_k)$$

In practice the last step isn't very easy to do so just do the following:

- Compute $T(r)$ and $G(z)$ tables and find pairs $T(r_i) \rightarrow G(z_j)$, that are the ones with closest value.
- Now the final PDF is obtained by plotting the PDF value of $T(r_i)$ in the intensity q of $G(z_q)$. If more than one $T(r_j)$ is assigned to the same $G(z_q)$ then just sum the PDF values and the mapping is obtained by checking what $s_k \rightarrow z_q$, that is $k \rightarrow q$.

We use a target function as a flat histogram isn't necessarily good (ex. night photo must be dark)

Example.

Suppose to have the following image PDF

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

PDF

Now, consider also the following target PDF

z_q	Specified $p_z(z_q)$
$z_0 = 0$	0.00
$z_1 = 1$	0.00
$z_2 = 2$	0.00
$z_3 = 3$	0.15
$z_4 = 4$	0.20
$z_5 = 5$	0.30
$z_6 = 6$	0.20
$z_7 = 7$	0.15

Target PDF

By Calculating $G(z_q)$ and the CDF $T(r_k)$ it is possible to now link s and z

s_i	$T(r_i)$	rounded	$G(z_q)$	value	rounded	S_i	Z_q
S_0	1.33	1	$G(z_0)$	0	0	S_0	Z_3
S_1	3.08	3	$G(z_1)$	0	0	S_1	Z_4
S_2	4.55	5	$G(z_2)$	0	0	S_2	Z_5
S_3	5.67	6	$G(z_3)$	1.05	1	S_3	Z_6
S_4	6.23	6	$G(z_4)$	2.45	2	S_4	Z_6
S_5	6.65	7	$G(z_5)$	4.55	5	S_5	Z_7
S_6	6.86	7	$G(z_6)$	5.95	6	S_6	Z_7
S_7	7.00	7	$G(z_7)$	7.00	7	S_7	Z_7

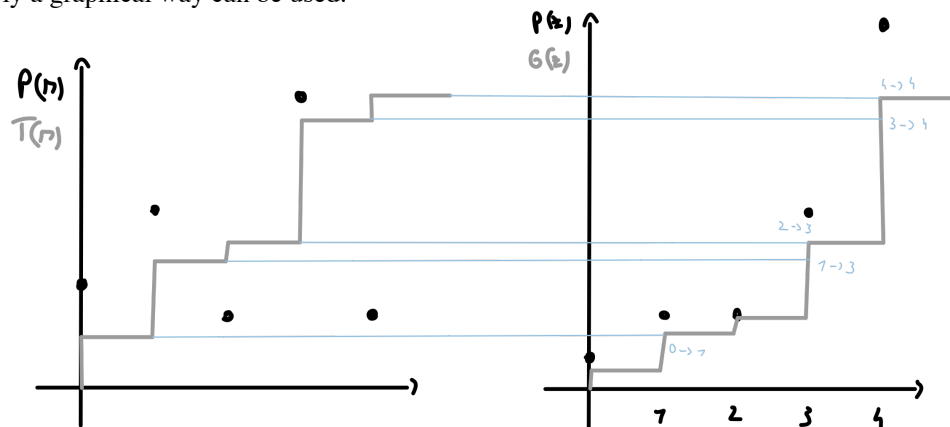
Final Tables

And now

$p_z(z_0)$	$p_z(z_1)$	$p_z(z_2)$	$p_z(z_3)$	$p_z(z_4)$	$p_z(z_5)$	$p_z(z_6)$	$p_z(z_7)$
0	0	0	$s_0 = 0.19$	$s_1 = 0.25$	$s_2 = 0.21$	$s_3 + s_4 = 0.24$	$s_5 + s_6 + s_7 = 0.11$

Remark.

Alternatively a graphical way can be used:



Example